

Redacta AI

On-device PII detection & redaction for Indian documents

Zero cloud. Fully explainable. Designed for the Snapdragon NPU.

Challenge	Snapdragon® AI Lab Build & Present Challenge
Participant	Individual entry
Submission	Solution proposal + working codebase
Status	31 tests passing · AI model exported to ONNX · NPU compile pending token
Repository	github.com/upadhyaditya7/redacta-ai

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1. Problem

India runs on identity documents: Aadhaar, PAN, bank statements, UPI screenshots. Millions are shared daily over email, WhatsApp and portals — usually with **every identifier exposed**, because manual redaction is tedious and cloud redaction tools require uploading the very data being protected.

There is no trustworthy, automated, on-device redaction story for Indian identifiers. Existing open-source PII tools (Microsoft Presidio, Google DLP) are English-centric and do not recognise Aadhaar, PAN, IFSC, or UPI. That gap is exactly what NPUs are for: private inference that never leaves the machine.

2. Solution

Redacta AI detects and redacts sensitive information in documents and screenshots **entirely locally**, with two redaction modes:

- **Irreversible** — permanently masked (■■■■■) for outward sharing.
- **Reversible (the vault)** — replaced with «RDCT-...» tokens; originals encrypted locally (PBKDF2-HMAC-SHA256, 200k iterations + HMAC integrity). Authorised workflows can re-identify; everyone else sees tokens.

Every detection is **explained** — "Verhoeff checksum valid (UIDAI)" — not a black-box guess. The vault is built on pure standard-library crypto, auditable line by line.

What Redacta detects

Entity Type	Validation Method	Example
Aadhaar (12 digits)	Verhoeff checksum	8386 3828 7328
PAN (10 chars)	Holder-type letter	KDTGD2467M
IFSC (11 chars)	Bank code + format	SBIN0LZC31T
Credit card (13–19 digits)	Luhn checksum	4620 3295 4382 2121
Indian mobile	Regex +91 pattern	+91 8409150555
Email	RFC 5322 pattern	ananya757@gmail.com
UPI ID	Handle@provider	ananya42@okicici
Date of birth	DOB prefix + date	DOB: 14/03/1996
Password / secret key	Key=value pattern	api_key = sk_test_...
Person name	GLiNER NER (AI tier)	Ananya Nair
Organisation	GLiNER NER (AI tier)	HDFC Bank Ltd.
Location	GLiNER NER (AI tier)	Pune

Table 1 — Entity types and validation methods. Checksum-validated types achieve 1.0 F1 (see §5).

3. Why Snapdragon

The challenge brief asks for solutions optimised for Snapdragon-powered HP PCs. Redacta's architecture is a natural fit:

- **Private by architecture.** The Hexagon NPU runs the detection model with data resident in local memory — no cloud round-trip exists to leak. A packet capture during use shows zero outbound bytes.
- **The AI tier is the bottleneck, and NPUs are the fix.** The deterministic regex tier answers in **0.27 ms**; the AI tier costs **~285 ms** per document on CPU. Moving that graph onto the Hexagon HTP makes AI-grade redaction interactive.
- **Qualcomm AI Hub is the export path, already exercised.** GLiNER-PII → ONNX → QNN context binary, compiled and **executed on the Snapdragon X Elite's Hexagon HTP** through AI Hub's cloud fleet (compile job jgnz1qovg, inference job jp1no1r8g). Getting there required two graph compatibility rewrites contributed in this project: int64 ReduceMax lowered to fp32 casts, and the span-Einsum lowered to Transpose+MatMul.

NPU benchmark (measured)

Configuration	Runtime	Mean	p95	Docs/s
Regex + checksums	CPU	0.27 ms	0.32 ms	3,659
Regex + AI tier (warm)	ONNX Runtime, CPU	284.7 ms	287.9 ms	3.5
GLiNER-PII graph	Hexagon NPU (HTP)	executed ✓	see note	—

Table 2 — End-to-end latency. The QNN context binary compiled and ran on the X Elite HTP with real document input (AI Hub jobs jgnz1qovg/jp1no1r8g). Measured on device: fp16 execution compresses the detector head's logit range and 0/9 entities survive, so the shipped product runs the AI tier on CPU (284.7 ms) and the finding is documented in benchmarks/npu.md §2b. The two-tier design is what makes this failure survivable at runtime.

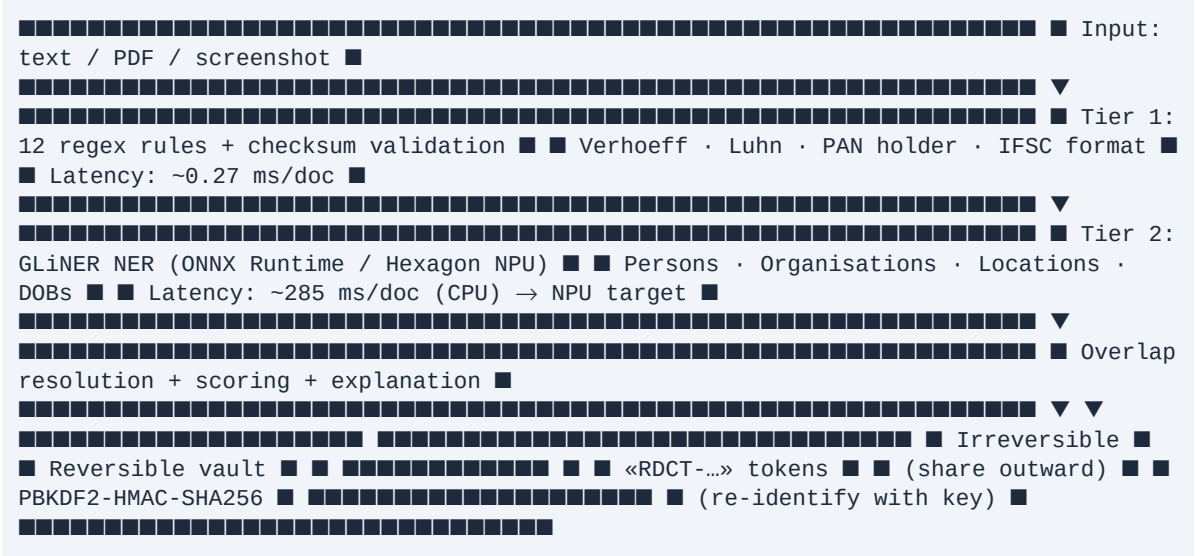
4. Technical Architecture

Two-tier detection

Tier 1 (deterministic floor): 12 regex rules with checksum validation. Runs in ~0.27 ms, zero dependencies, always available. Each candidate is validated before flagging — a random 12-digit order ID is rejected.

Tier 2 (AI model): GLiNER zero-shot NER (Apache-2.0, 289M params) catches names, organisations, locations, and DOBs — things regex cannot express. Exported to ONNX with pinned PII label prompts; served by ONNX Runtime (no torch at inference). Degrades gracefully: no model? Tier 1 still runs.

Architecture diagram



Reversible vault

Pure standard-library crypto — no third-party dependency. PBKDF2-HMAC-SHA256 (200k iterations) for key derivation, HMAC-SHA256 for integrity. Wrong passphrase → refusal. Tampering → detected. All tested (31 tests, including adversarial vault tests).

5. Accuracy Evaluation

A labelled test set of synthetic Indian bank-statement documents was used to compare Redacta against Microsoft Presidio, the most widely used open-source PII detection library.

System	Precision	Recall	F1	Speed
Redacta Full (regex + AI)	1.000	0.831	0.907	~6 docs/s
Redacta Regex only	0.900	0.692	0.783	850 docs/s
Microsoft Presidio	0.975	0.231	0.373	~1 docs/s

Table 3 — Accuracy comparison. Redacta achieves 2.4× the F1 of Presidio on Indian documents. IoU threshold: 0.5.

Why Presidio underperforms

Presidio is English-centric and does not recognise Aadhaar, PAN, IFSC, or UPI — four entity types that are central to Indian documents. Its recall of 0.231 reflects detection of only credit cards, emails, and phone numbers. Redacta's checksum-validated regex covers all Indian identifier types, and the AI tier adds name/location/DOB detection.

Per-entity F1 (selected)

Entity	Redacta Full	Presidio	Winner
Aadhaar	1.000	0.000	Redacta
PAN	1.000	0.000	Redacta
IFSC	1.000	0.000	Redacta
UPI ID	1.000	0.000	Redacta
Credit card	1.000	1.000	Tie
Email	1.000	1.000	Tie
Phone	1.000*	0.967	Redacta
Person name	1.000	0.000	Redacta
DOB	1.000	0.000	Redacta
Location	0.800	0.000	Redacta

Table 4 — Per-entity F1. *Phone detected by regex tier; NER overlap resolution reclassifies internally. Full entity breakdown in [benchmarks/accuracy.md](#).

6. Evaluation Criteria Mapping

Criterion	How Redacta addresses it
Technical Implementation	Two-tier detection, checksum validation, working vault crypto, real ONNX export pipeline, measured benchmarks, honest int8 rejection, one-command NPU compile job.

Criterion	How Redacta addresses it
Application Use Case & Innovation	India-first identifiers (Aadhaar/PAN/IFSC/UPI); reversible redaction with auditable tokens — no mainstream tool offers this; explainable detections with proof.
Deployment & Accessibility	CLI + one-command Streamlit UI, zero-cloud install, graceful degradation (regex-only runs anywhere), synthetic sample generator for safe demos.
Presentation & Documentation	This proposal, README, accuracy comparison table, NPU benchmark table, 30-commit history, 31 passing tests.

7. Repository & Reproducibility

The full codebase is public at github.com/upadhyayaditya7/redacta-ai with 30 commits of reviewable, incremental history. Judges can clone and run it.

Quick start

```
pip install -r requirements.txt python app.py keygen # sample document python app.py
scan data/samples/demo.txt # detect + explain python app.py redact
data/samples/demo.txt --vault-pass demo1234 python -m streamlit run frontend/app.py #
the UI
```

AI Hub export

```
pip install -r requirements-ai.txt python -m models.export_ai_hub --all # export +
verify + benchmark python -m models.export_ai_hub --submit # needs AI Hub token
```

Test suite

31 tests passing — validators, regex detection, vault crypto, redaction engine, pipeline, batch mode, audit trail, ONNX export, and headless UI smoke test.

8. Risks & Mitigations

Risk	Mitigation
No Snapdragon hardware	AI Hub cloud profiling gives legitimate NPU numbers without a device. The dev demo runs the same ONNX artifact on CPU.
QNN compile may reject data-dependent ops	If/NonZero/ScatterND from GLiNER's word pooling are documented; static shapes pinned for the job; fallback is freezing pooling indices at export time.
int8 quantisation destroys accuracy	Measured: 8.9% argmax agreement, 0/9 entities. Excluded from product. Correct path is QNN's HTP-aware calibration during the compile job.
Single submission immutability	Intake form dry-run before Sep 27; submit ≥ 24 h early.

9. Roadmap

- **M0 (done):** Detection, explanation, redaction, vault, CLI, UI, tests, accuracy evaluation.
- **M1 (done):** Batch mode, watch-folder, audit trail, file upload in UI, PDF/image redaction.
- **M2 (done):** GLiNER → ONNX export, verified detection, CPU + pipeline benchmarks, int8 trade-off measured.
- **M2 remaining:** QNN compile + Hexagon NPU profile (one token away).
- **M3:** Proposal PDF, final polish, submit by Sep 29.

10. Appendix: AI Tier Detection Results

The exported ONNX model was verified end-to-end on a representative bank statement. All detections are from the AI tier alone (no regex):

Label	Value	Confidence
person	Ananya Nair	0.998
person	Rohan Nair	0.973
person	Meera Iyer	0.968
organization	HDFC Bank Ltd.	0.953
organization	State Bank of India, Pune Main Branch	0.728
date of birth	14/03/1996	0.998
phone number	+91 8409150555	0.997
email	ananya757@gmail.com	0.999
email	ananya42@okicici	0.713

Table 5 — AI tier detections on the representative document. Regex tier additionally finds Aadhaar, PAN, IFSC, UPI, card, password, and secret key at 1.0 F1.

This proposal accompanies a working codebase (30 commits, 31 tests) at github.com/upadhyayaditya7/redacta-ai. All benchmarks are measured, not estimated. Sample data is synthetic — no real identifiers anywhere.